

Tarea Sistemas Lineales

- ① Encontrar la solución general para cada uno de los sistemas diferenciales lineales

$$a) \frac{d}{dt} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Diagonalización

$$\det(A - \lambda I) = 0$$

$$\begin{bmatrix} 2-\lambda & 0 & 0 \\ 1 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{bmatrix} = \begin{matrix} (2-\lambda) [(1-\lambda)^2 - 0] - \\ (0) [(1-\lambda) - 0] + \\ (0) [0 - (1-\lambda)] \end{matrix}$$

$$= (2-\lambda)(1-\lambda)(1-\lambda) = 0$$

$$\lambda = 2$$

$$\lambda = 1$$

$$\lambda = 1$$

E₂

$$\begin{bmatrix} -2+2 & 0 & 0 \\ -1 & -1+2 & 0 \\ 1 & 0 & -1+2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} u &= u \\ -u + v &= 0, \quad v = u \\ u + w &= 0, \quad w = -u \end{aligned}$$

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} u \\ u \\ -u \end{pmatrix} = u \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$V_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

E₂

$$\begin{bmatrix} -2 & 0 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-u = 0$$

$$-u = 0$$

$$u = 0$$

$$V_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

v y w parámetros
libres

$$V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Soluciones Particulares

$$x_1 = e^{2t} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$x_2 = e^t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$x_3 = e^t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Matrix fundamental

$$\Phi(t) = \begin{bmatrix} -e^{2t} & 0 & 0 \\ -e^{2t} & e^t & 0 \\ -e^{2t} & 0 & e^t \end{bmatrix} \quad \begin{aligned} & \frac{d}{dt} \\ & = -e^{2t} (e^t \cdot e^t) \\ & = -e^{2t} e^{2t} \\ & = -e^{4t} \neq 0 \end{aligned}$$

Solution General

$$X(t) = \begin{bmatrix} a e^{2t} & 0 & 0 \\ -a e^{2t} & b e^t & 0 \\ -a e^{2t} & 0 & c e^t \end{bmatrix} \quad \begin{aligned} & a, b, c \in \mathbb{R} \\ & \det = -e^{2t} \end{aligned}$$

$$b) \quad \frac{d}{dt} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Solution Particular

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$X_p = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3/2 \\ 1/2 \end{bmatrix}$$

$$\begin{aligned} \hat{x} &= -1 \\ \hat{y} &= -3/2 \\ \hat{z} &= 1/2 \end{aligned}$$

Matr. Eigenwerte

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} \lambda - 2 & 0 & 0 \\ 1 & \lambda - 1 & 0 \\ -1 & 0 & \lambda - 1 \end{bmatrix} = \det \begin{bmatrix} \lambda - 2 & 0 & 0 \\ 1 & \lambda - 1 & 0 \\ -1 & 0 & \lambda - 1 \end{bmatrix} = (\lambda - 2) [(\lambda - 1)^2] -$$

$$(-1) [(\lambda - 1)]$$

$$(-1) [(\lambda - 1)]$$

$$= \lambda = 2 \quad \lambda = 1$$

$$\lambda = 1$$

E_2

$$\begin{bmatrix} -2+2 & 0 & 0 \\ -1 & -1+2 & 0 \\ 1 & 0 & -1+2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u = u$$

$$-u + v = 0$$

$$v = u$$

$$u + w = 0$$

$$w = -u$$

$$V_1 = \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} u \\ u \\ -u \end{pmatrix} = u \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad V_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

E₁

$$\begin{bmatrix} -2 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 \\ 1 & 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -u &= 0 \\ -u &= 0 \\ u &= 0 \end{aligned}$$

$$v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Sollicitous vectors

$$x_1 = e^{2t} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad x_2 = e^t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad x_3 = e^t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solution General

$$x(t) = \begin{bmatrix} e^{2t} & 0 & 0 \\ e^{2t} & e^t & 0 \\ -e^{2t} & 0 & e^t \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} -1 \\ -3/2 \\ 1/2 \end{bmatrix}$$

Φ C x_p

$$x(t) = \begin{bmatrix} a e^{2t} \\ a e^{2t} + b e^t \\ -a e^{2t} + c e^t \end{bmatrix} + \begin{bmatrix} -1 \\ -3/2 \\ 1/2 \end{bmatrix}$$

$$c) \frac{d}{dt} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & -3 & 3 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

determinante

$$\begin{bmatrix} 2-\lambda & -3 & 3 \\ 0 & 3-\lambda & -2 \\ 0 & -1 & 2-\lambda \end{bmatrix} \quad \begin{aligned} (2-\lambda) [(3-\lambda)(2-\lambda) - 2] &= \\ (-3) [0] & \\ (3) [0] & \end{aligned}$$

$$(2-\lambda) [4 - 5\lambda + \lambda^2 - 2] = 0$$

$$(2-\lambda) (4 - 5\lambda + \lambda^2)$$

$$(2-\lambda) ((\lambda-4)(\lambda-1)) = 0$$

$$\lambda = 2$$

$$\lambda = 1$$

$$\lambda = 4$$

E_2

$$\begin{bmatrix} -2 & 2 & 3 & -3 \\ 0 & -3 & 2 & 2 \\ 0 & 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} 0 & 3 & -3 & 1 & 0 \\ 0 & -1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix} \quad F_3 = F_2 + F_3$$

$$\begin{pmatrix} 0 & 3 & -3 & 1 & 0 \\ 0 & -1 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 & 0 \end{pmatrix}$$

$$3v - 3w = 0$$

$$v = w$$

$$-v + 2w = 0$$

$$-v = -2w$$

$$v = 2w$$

$$v = 0$$

$$2w = 0$$

$$w = 0$$

$$V_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

E₁

$$\begin{bmatrix} -1 & 3 & -3 & 1 & 0 \\ 0 & -2 & 2 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \end{bmatrix} F_3 = \frac{1}{2} F_2 + F_3$$

$$\begin{matrix} u & v & w \\ \begin{bmatrix} -1 & 3 & -3 & 1 & 0 \\ 0 & -2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$W = W$$

$$-2V = -2W$$

$$V = W$$

$$-u + 3v - 3w = 0$$

$$-u = -3w + 3w$$

$$u = 0$$

$$V_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

E₂

$$\begin{bmatrix} 2 & 3 & -3 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 \end{bmatrix} F_3 = -F_2 + F_3$$

$$\begin{matrix} u & v & w \\ \begin{bmatrix} 2 & 3 & -3 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$W = W$$

$$V = -2W$$

$$2u + 3v - 3w = 0$$

$$2u = -3(-2w) + 3w$$

$$2u = 6w + 3w$$

$$u = \frac{9}{2}w$$

$$V_3 = \begin{pmatrix} 9/2 \\ -2 \\ 1 \end{pmatrix}$$

Soliciates vectors

$$x_1 = e^{2t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$x_2 = e^t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$x_3 = e^{4t} \begin{pmatrix} 9/2 \\ -2 \\ 1 \end{pmatrix}$$

Solution General

$$x(t) = \begin{pmatrix} a e^{2t} & 0 & c \frac{9}{2} e^{4t} \\ 0 & b e^t & -2c e^{4t} \\ 0 & b e^t & c e^{4t} \end{pmatrix}$$

$$a, b \in \mathbb{R}$$

$$\Phi \cdot C$$

② Encuentre la única solución con condición inicial

$$\begin{bmatrix} x(0) \\ y(0) \\ z(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ para } b \text{ y } c \text{ del ejercicio anterior}$$

Para PVI en b

$$\begin{bmatrix} -a & 0 & 0 \\ -a & b & 0 \\ a & 0 & c \end{bmatrix} + \begin{bmatrix} -1 \\ -3/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -a & 0 & 0 \\ -a & b & 0 \\ a & 0 & c \end{bmatrix} = \begin{bmatrix} 2 \\ 3/2 \\ -1/2 \end{bmatrix}$$

$$a = -2$$

$$-a + b = 3/2$$

$$a + c = -1/2$$

$$b = 3/2 - 1 = -1/2$$

$$c = -1/2 + 2 = 3/2$$

$$R11 \quad \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -2 \\ -1/2 \\ 3/2 \end{bmatrix}$$

(3) Resuelva $\frac{dx}{dt} = x + 2y$, $\frac{dy}{dt} = 2x + y - 2$

para $x(0) = 0$, $y(0) = 0$

$$x' = x + 2y + 0$$

$$y' = 2x + y - 2$$

$$\begin{bmatrix} x(0) \\ y(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

Solución Particular

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\det = -3$$

$$(1 \cdot 1) - (2 \cdot 2) = -3$$

$$= -\frac{1}{3} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= -\frac{1}{3} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$x_p = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix}$$

$$\hat{x} = -4/3$$

$$\hat{y} = -2/3$$

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 1 \end{bmatrix} = (\lambda - 1)^2 - 4 = \lambda^2 - 2\lambda + 1 - 4 \\ = \lambda^2 - 2\lambda - 3 \\ = (\lambda - 3)(\lambda + 1)$$

$$\lambda = 3$$

$$\lambda = -1$$

E_{-1}

$$\begin{bmatrix} -1 & -1 & -2 \\ -2 & -1 & -1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$F_2 = -F_1 + F_1 \rightarrow \begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-u - v = 0$$

$$v = -v$$

$$-u = v$$

$$V_n = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} -v \\ v \end{bmatrix} = v \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$V_n = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

E_2

$$\begin{bmatrix} -1+3 & -1 \\ -2 & -1+3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} F_2 = F_2 + F_1 \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$u - v = 0 \quad v = v$$

$$u = v$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} v \\ v \end{bmatrix} = v \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$V_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Soluciones Partes

$$x_1 = e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_2 = e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Matriz fundamental

$$\Phi = \begin{bmatrix} e^{3t} & -e^{-t} \\ e^{3t} & e^{-t} \end{bmatrix}$$

Solución General

$$x(t) = \begin{bmatrix} a e^{3t} & -b e^{-t} \\ a e^{2t} & b e^{-t} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix}$$

$$x(t) = \begin{bmatrix} a e^{3t} & -b e^{-t} \\ a e^{2t} & b e^{-t} \end{bmatrix} + \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix} \quad a, b \in \mathbb{R}$$

⑤ Encontrar la solución General del sistema

La Matriz fundamental

La Matriz fundamental principal en $t=0$

$$a) \quad \frac{d}{dt} x = \begin{bmatrix} 3 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 2 \end{bmatrix} x, \quad x' = Ax$$

$$\begin{bmatrix} 3-\lambda & 0 & 0 \\ 1 & -2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{bmatrix} = (3-\lambda) [-(2+\lambda)(2-\lambda)] - \\ 0 [(2-\lambda)-0] + \\ 0 [0]$$

$$= (3-\lambda)(-2-\lambda)(2-\lambda) = 0$$

$$\lambda = 3$$

$$\lambda = -2$$

$$\lambda = 2$$

$$\begin{array}{rcl} -300 & = & 0-60 \\ 3-60 & & \end{array}$$

$$-u =$$

$$\begin{bmatrix} -3 & 0 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 5 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-u + 5v = 0 \quad v = t \quad w = 0$$

$$u = 5v$$

$$u = 5t$$

$$V_1 = \begin{pmatrix} 5t \\ t \\ 0 \end{pmatrix} = t \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$$

$$F_2 = -F_1 + F_2$$

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$u = 0$$

$$w = t$$

$$v = 0$$

$$V_2 = \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} t$$

E_2

$$\begin{bmatrix} -5 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -4 \end{bmatrix} \quad F_2 = 5F_2 - F_1 \quad \begin{bmatrix} -5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -4 \end{bmatrix} F_3 \leftrightarrow F_1$$

$$\begin{bmatrix} -5 & 0 & 0 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} F_1 \div -5 \\ F_2 \div -4 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u = 0 \quad v = t \quad w = 0$$

$$V_3 = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Soluciones Particulares y Matriz Fundamental

$$x_1 = e^{3t} \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix} \quad x_2 = e^{2t} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad x_3 = e^{-2t} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\Phi = \begin{bmatrix} 5e^{3t} & 0 & 0 \\ e^{3t} & 0 & e^{-2t} \\ 0 & e^{2t} & 0 \end{bmatrix}$$

Sol: Lösung General

$$x(t) = \begin{bmatrix} a_5 e^{3t} & 0 & 0 \\ a e^{3t} & 0 & c e^{-2t} \\ 0 & b e^{2t} & 0 \end{bmatrix}$$

M, f. für P.V.I (t=0)

$$x(0) = \begin{bmatrix} a_5 & 0 & 0 \\ a & 0 & c \\ 0 & b & 0 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{\Phi}$

$$b) \frac{d}{dt} x = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} x$$

$$\begin{bmatrix} 2-\lambda & 0 & 0 \\ 1 & 1-\lambda & 0 \\ -1 & 0 & 1-\lambda \end{bmatrix} = \det$$

$$= (2-\lambda) [(1-\lambda)^2]$$

$$\lambda = 2 \quad \lambda = 1$$

$$\lambda = 1$$

E_2

$$\begin{bmatrix} -2 & 0 & 0 \\ -1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$F_2 \leftrightarrow F_3 \quad \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} F_1 = F_1 + F_2 \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u + w = 0 \quad v + w = 0 \quad w = w$$
$$u = -w \quad v = -w$$

$$V_1 = \begin{pmatrix} -w \\ -w \\ w \end{pmatrix} = w \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

E_1

$$\begin{bmatrix} -2+1 & 0 & 0 \\ -1 & -1+1 & 0 \\ 1 & 0 & -1+1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$F_2 = F_2 + F_3, \quad F_3 = F_2 + F_1, \quad F_1 = -F_1$$

$$u = 0$$

$$v = v$$

$$w = w$$

$$V = \begin{bmatrix} 0 \\ v \\ w \end{bmatrix} = v \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + w \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Soluciones rectas

$$x_1 = e^{2t} \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \quad x_2 = e^t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad x_3 = e^t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Matriz fundamental

$$\Phi = \begin{bmatrix} -e^{2t} & 0 & 0 \\ -e^{2t} & e^t & 0 \\ e^{2t} & 0 & e^t \end{bmatrix}$$

Solución General

$$x(t) = \begin{bmatrix} -e^{2t} & 0 & 0 \\ -e^{2t} & e^t & 0 \\ e^{2t} & 0 & e^t \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$x(t) = \begin{bmatrix} -ae^{2t} & 0 & 0 \\ -ae^{2t} & be^t & 0 \\ ae^{2t} & 0 & ce^t \end{bmatrix}$$

M.F. para $t=0$

$$X(0) = \begin{bmatrix} -a & 0 & 0 \\ -a & b & 0 \\ a & 0 & c \end{bmatrix}$$

$$c) \frac{d}{dt} X = \begin{bmatrix} 2 & -3 & 3 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix} X$$

$$\begin{bmatrix} 2-\lambda & -3 & 3 \\ 0 & 3-\lambda & -2 \\ 0 & -1 & 2-\lambda \end{bmatrix} \quad \text{det}$$
$$(2-\lambda)[(3-\lambda)(2-\lambda)-2] -$$
$$(-3)[0-0] +$$
$$(3)[0-0]$$

$$= (2-\lambda)[6-5\lambda+\lambda^2-2]$$

$$= (2-\lambda)[4-5\lambda+\lambda^2]$$

$$\lambda = 2$$

$$\lambda = 1$$

$$\lambda = 4$$

E_4

$$\begin{bmatrix} 2 & 3 & -3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{3}{2} & -\frac{3}{2} \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -9/2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -9/2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u - 9/2 v = 0$$

$$v = -2w$$

$$w = w$$

$$u = \frac{9}{2} v$$

$$v = -2t$$

$$w = t$$

$$u = \frac{9}{2} t$$

$$V_1 = \begin{bmatrix} 9/2 \\ -2 \\ 1 \end{bmatrix} t$$

E_2

$$\begin{bmatrix} 0 & -3 & 3 \\ 0 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u = t$$

$$v = 0$$

$$w = 0$$

$$V_2 = t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

E₁

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & 2 & -2 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u = 0$$

$$v - w = 0$$

$$w = t$$

$$v = w$$

$$v = t$$

$$V_3 = t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Soluciones vecdas y matriz fundamental

$$x_1 = e^{4t} \begin{pmatrix} 9/2 \\ -2 \\ 1 \end{pmatrix} \quad x_2 = e^{2t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad x_3 = e^t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\Phi = \begin{bmatrix} 9/2 e^{4t} & e^{2t} & 0 \\ -2 e^{4t} & 0 & e^t \\ e^{4t} & 0 & e^t \end{bmatrix}$$

Solucion General

$$x(t) = \begin{bmatrix} 9/2 a e^{4t} & b e^{2t} & 0 \\ -2 a e^{4t} & 0 & c e^t \\ a e^{4t} & 0 & c e^t \end{bmatrix}$$

M.F Para $t=0$

$$x(0) = \begin{bmatrix} a/2 & a & b & 0 \\ -2a & 0 & c & \\ a & 0 & c & \end{bmatrix}$$

b) Encontrar solución única con $\begin{bmatrix} x(0) \\ y(0) \\ z(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

en b)

$$\begin{bmatrix} -a & 0 & 0 \\ -a & b & 0 \\ a & 0 & c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$-a = 1, \quad a = -1$$

$$1 + b = 0, \quad b = -1$$

$$-1 + c = 0, \quad c = 1$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

en c

$$\begin{bmatrix} \frac{a}{2} & b & 0 \\ -2a & 0 & c \\ a & 0 & c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{cases} \frac{a}{2} + b = 1 \\ -2a + c = 2 \\ a + c = 1 \end{cases}$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -1/3 \\ 5/2 \\ 4/3 \end{bmatrix}$$

7) Resuelve los sig. problemas de valor inicial

$$a) \begin{cases} \frac{d}{dt} x = x + 2y \\ \frac{d}{dt} y = 2x + y - 2 \end{cases}, \quad x(0) = y(0) = 0$$

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

Solución Particular

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\begin{aligned} \det & \\ (1,1) - (2,2) & \\ = -3 & \end{aligned}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= -\frac{1}{3} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= -\frac{1}{3} \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$x_p = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix}$$

$$\tilde{x} = -4/3$$

$$\tilde{y} = -2/3$$

Matriz fundamental

$$\Phi = \begin{bmatrix} e^{3t} & -e^{-t} \\ e^{3t} & e^{-t} \end{bmatrix}$$

Solución General

$$x(t) = \begin{bmatrix} ae^{3t} & -be^{-t} \\ ae^{3t} & be^{-t} \end{bmatrix} + \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix}$$

P.V.2.

$$x(0) = y(0) = 0$$

$$\begin{bmatrix} a & -b \\ a & b \end{bmatrix} + \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} a & -b \\ a & b \end{bmatrix} = \begin{bmatrix} -4/3 \\ -2/3 \end{bmatrix}$$

$$F_2 = F_2 - F_1$$

$$\begin{bmatrix} a & -b \\ 0 & 2b \end{bmatrix} = \begin{bmatrix} -4/3 \\ 2/3 \end{bmatrix} \div 2$$

$$\begin{bmatrix} a & -b \\ 0 & b \end{bmatrix} = \begin{bmatrix} -4/3 \\ 1/3 \end{bmatrix}$$

$$F_1 = F_1 + F_2$$

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} -3/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1/3 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -1 \\ 1/3 \end{bmatrix}$$

$$6) \frac{d}{dt} x = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} x + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2-\lambda & 0 & 0 \\ 1 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

$$(2-\lambda)(1-\lambda)(1-\lambda)$$

$$\lambda = 2$$

$$\lambda = 1$$

E_2

$$\begin{bmatrix} -2 & 2 & 0 & 0 \\ -1 & -1 & 2 & 0 \\ 1 & 0 & 0 & -1+2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$u = u$$

$$-u + v = 0 \\ v = u$$

$$u + w = 0 \\ w = -u$$

$$V_1 = \begin{pmatrix} u \\ u \\ -u \end{pmatrix} = u \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

E_3

$$\begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-u = 0 \\ u = 0$$

$$v = t$$

$$w = s$$

$$V_0 = \begin{bmatrix} 0 \\ t \\ s \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

X → Soluciones Vectoriales

$$x_1 = e^{2t} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad x_2 = e^t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad x_3 = e^t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Matriz Fundamental

$$\Phi = \begin{bmatrix} e^{2t} & 0 & 0 \\ e^{2t} & e^t & 0 \\ e^{2t} & 0 & e^t \end{bmatrix}$$

Solución homogénea

$$x_h(t) = a e^{2t} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + b e^t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c e^t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$x_h(t) = \begin{pmatrix} a e^{2t} & 0 & 0 \\ a e^{2t} & b e^t & 0 \\ -a e^{2t} & 0 & c e^t \end{pmatrix}$$

Solución Particular

$$x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 0 \\ -1/2 & 1 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix}$$

Solution General

$$X(t) = \begin{bmatrix} a e^{2t} & 0 & 0 \\ a e^{2t} & b e^t & 0 \\ -a e^{2t} & 0 & c e^t \end{bmatrix} + \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix}$$

P.V.I. $x(0) = y(0) = 0$

$$\begin{bmatrix} a & 0 & 0 \\ a & b & 0 \\ -a & 0 & c \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

$$a = 2$$

$$-2 + c = -1$$

$$2 + b = 0$$

$$c = 1$$

$$b = -2$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$X(t) = \begin{bmatrix} 2 e^{2t} & 0 & 0 \\ 2 e^{2t} & -2 e^t & 0 \\ -2 e^{2t} & 0 & e^t \end{bmatrix} + \begin{bmatrix} -2 \\ -3 \\ 1 \end{bmatrix}$$

$$c) \frac{dx}{dt} = -3x + y + 3t$$

$$x(0) = y(0) = 0$$

$$\frac{dy}{dt} = 2x - 4y + t e^{-t}$$

$$\begin{bmatrix} -3 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 3t \\ t e^{-t} \end{bmatrix}$$

$$f(t) = \begin{pmatrix} 3t \\ t e^{-t} \end{pmatrix}$$

det

$$(-3)(-4) - [2] \quad \begin{bmatrix} -3-\lambda & 1 \\ 2 & -4-\lambda \end{bmatrix}$$

$$= 12 - 2$$

$$= 10$$

$$(-3-\lambda)(-4-\lambda) - [2]$$

$$(3+\lambda)(4+\lambda) - 2$$

$$12 + 7\lambda + \lambda^2 - 2$$

$$10 + 7\lambda + \lambda^2$$

$$(5+\lambda)(2+\lambda) = 0$$

$$\lambda = -2$$

$$\lambda = -5$$

E-2

$$\begin{bmatrix} 3-2 & -1 \\ -2 & 4-2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u - v = 0$$

$$v = 1$$

$$u = v$$

$$u = 1$$

$\lambda = -5$

$$\begin{bmatrix} 3-5 & -1 \\ -2 & 4-5 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2u + v = 0$$

$$v = -2u$$

$$u = 1$$

$$v = -2$$

$$V_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Soluciones rectas

$$x_1 = e^{-2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad x_2 = e^{-5t} \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Matriz Fundamental

$$\Phi = \begin{bmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{bmatrix}$$

Solución Particular

$$x_p = \Phi(t) \int \Phi^{-1}(t) f(t) dt$$

$\det \Phi$

$$(e^{-2t})(-2e^{-5t}) - (e^{-2t})(e^{-5t})$$

$$= -2e^{-7t} - e^{-7t}$$

$$\Phi^{-1} = \frac{1}{-3e^{-7t}} \begin{pmatrix} -2e^{-5t} & -e^{-5t} \\ -e^{-2t} & e^{-2t} \end{pmatrix} = -3e^{-7t}$$